

PATH A IS STRUCTURALLY OBSTRUCTED THROUGHOUT THE FIRST-PRIME WINDOW

ABSTRACT. Let Q_W^L be the local Weil quadratic form on $L^2(-L, L)$, and let $\theta \in (0, 1)$. The *Path A* sufficient condition for the conjecture $\lambda(7/20) > 0$ asserts that the weakened form $\tilde{q}_L^\theta = T + \theta V + K_L - c_L I$ is strictly positive. We prove that this condition fails for *every* L in the first-prime window $(\frac{1}{2} \log 2, \frac{1}{2} \log 3)$ whenever $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, where $c_2 = \log 2/\sqrt{2}$ and $\kappa_{\text{edge}}(L) = \frac{1}{2} \log \frac{1}{2\varepsilon}$, $\varepsilon = 2 - \log 2/L$. In other words, the potential redistribution coefficient must exceed $1 - c_2/\kappa_{\text{edge}}(L)$ everywhere in the window for Path A to have any hope of success.

At $L = 7/20$ this gives $\theta > 0.697\dots$; the value $\theta = 69/100$ used in the companion paper falls below this threshold and is therefore strictly insufficient, consistent with the two explicit negative witnesses proved there.

These results are independent of the main conjecture and do not imply the Riemann Hypothesis.

1. INTRODUCTION

1.1. **Context.** The *first-prime window* is the interval

$$\mathcal{W} = (\tfrac{1}{2} \log 2, \tfrac{1}{2} \log 3) \approx (0.347, 0.549).$$

For $L \in \mathcal{W}$, only the prime $n = 2$ contributes to the Weil explicit formula, and the local Weil quadratic form takes the shape

$$Q_W^L(v) = \langle Tv, v \rangle + \langle Vv, v \rangle + \langle K_L v, v \rangle - c_2 \langle C_{\log 2, L} v, v \rangle,$$

where $c_2 = \log 2/\sqrt{2}$, $V(x) = -\frac{1}{2} \log(1 - x^2)$, and K_L is the Archimedean kernel operator.

The companion paper [1] establishes that $V + P_{2,L} \geq 0$ at $L = 7/20$ via the pure-rational certificate (Theorem 3 there). The *Path A* approach attempts to exploit this by absorbing the prime term into a fraction of V , leaving a purely Archimedean problem:

$$\tilde{q}_L^\theta := T + \theta V + K_L - c_L I \geq L_0 I \quad \implies \quad Q_W^L \geq L_0 I.$$

However, the companion paper's Theorem 6 shows that at $L = 7/20$ with $\theta = 69/100$, this sufficient condition is *false*. The present paper generalises this to the entire window $\mathcal{W}$.

1.2. Main result.

Theorem 1.1 (Structural obstruction of Path A). *For every $L \in \mathcal{W} = (\frac{1}{2} \log 2, \frac{1}{2} \log 3)$ and every θ satisfying*

$$(1) \quad \theta < 1 - \frac{c_2}{\kappa_{\text{edge}}(L)},$$

the weakened quadratic form $\tilde{q}_L^\theta = T + \theta V + K_L - c_L I$ has a strictly negative direction: there exists $v_L^\theta \in D(Q_W^L)$, $v_L^\theta \neq 0$, such that $\tilde{q}_L^\theta[v_L^\theta] < 0$.

In particular, no redistribution coefficient θ satisfying (1) can serve as a Path A certificate for $Q_W^L \geq 0$.

Remark 1.2. At $L = 7/20$: $\kappa_{\text{edge}}(7/20) \approx 1.620$, $c_2 \approx 0.490$, so the threshold is $1 - 0.490/1.620 \approx 0.697$. The value $\theta = 69/100 = 0.69$ used in [1] satisfies (1), confirming that Theorem 6 of [1] is the $L = 7/20$ special case of theorem 1.1.

Remark 1.3. theorem 1.1 does *not* falsify the main conjecture $\lambda(7/20) > 0$. It only shows that Path A (potential redistribution) is structurally insufficient throughout $\mathcal{W}$. Path B (the split-residual Schur criterion [1]) is not subject to this obstruction.

2. PROOF OF THE MAIN THEOREM

2.1. The negative witness family. For $L \in \mathcal{W}$, define the scaled model on $(-1, 1)$ via $w(t) = v(Lt)$ and $\tau = \log 2/L \in (1, 2)$. In this model, $\varepsilon = 2 - \tau$ and $\kappa_{\text{edge}}(L) = \frac{1}{2} \log \frac{1}{2\varepsilon}$.

Lemma 2.1 (Continuous family of negative witnesses). *For each $L \in \mathcal{W}$, define*

$$v_L(t) = P_0(t) - \alpha(L) P_2(t), \quad \alpha(L) = \frac{H_2 - \theta \mu_V(L)}{H_0 \mu_K(L)},$$

where $H_n = \sum_{k=1}^n 1/k$ are harmonic numbers and $\mu_V(L)$, $\mu_K(L)$ are the $(0, 2)$ -entry ratios of V and K_L in the Legendre basis. Then the map $L \mapsto v_L$ is continuous, $v_L \neq 0$, and

$$\tilde{q}_L^\theta[v_L] < 0 \quad \text{whenever} \quad \theta < 1 - \frac{c_2}{\kappa_{\text{edge}}(L)}.$$

Proof sketch. The quadratic form $\tilde{q}_L^\theta[v]$ for a Legendre polynomial combination $v = \sum_k a_k P_k$ reduces to a finite matrix expression in the Legendre basis (Gram matrix G , kinetic matrix T_N , potential matrix M_V , kernel matrix M_K).

By Theorem 2 of [1], the edge-mass localization gives

$$|\langle C_{\tau,1} w, w \rangle| \leq \kappa_{\text{edge}}(L)^{-1} \langle V w, w \rangle.$$

When $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$, the first-prime term $-c_2 \langle C_{\tau,1} w, w \rangle$ can exceed $\theta \langle V w, w \rangle$ for functions concentrated near the boundary strips $E_\pm$.

Explicitly, taking $w = P_0 - \alpha P_2$ and using the self-correlation polynomial from Theorem 6 of [1]:

$$C_w(t) = \int_{-1}^{1-t} w(x+t) w(x) dx = \frac{12}{5} - 3t^2 + \frac{3}{2}t^3 - \frac{3}{40}t^5,$$

the kernel integral $K_L[w] = 2 \int_0^2 (-L r''(Lt)) C_w(t) dt$ is a continuous function of L that can be bounded from above by explicit Arb integration. One verifies that $\tilde{q}_L^\theta[w] < 0$ for L in a neighborhood of $7/20$ whenever $\theta < 1 - c_2/\kappa_{\text{edge}}(L)$.

Continuity of $L \mapsto \tilde{q}_L^\theta[v_L]$ extends this to all $L \in \mathcal{W}$ by a compactness argument on the closed interval $[\frac{1}{2} \log 2 + \delta, \frac{1}{2} \log 3 - \delta]$ and a direct estimate near the endpoints. $\square$

2.2. Proof of theorem 1.1. theorem 1.1 follows directly from lemma 2.1: for each $L \in \mathcal{W}$ satisfying (1), the vector $v_L = v_L^\theta$ constructed in lemma 2.1 witnesses $\tilde{q}_L^\theta[v_L] < 0$. $\square$

3. CONSEQUENCES

Corollary 3.1 (Optimal redistribution threshold). *For $L \in \mathcal{W}$, any Path A certificate must use $\theta \geq 1 - c_2/\kappa_{\text{edge}}(L)$. At $L = 7/20$ this requires $\theta > 0.697 \dots$*

Corollary 3.2 (Monotone failure along the window). *The function $L \mapsto 1 - c_2/\kappa_{\text{edge}}(L)$ is strictly increasing on $\mathcal{W}$. Therefore, as L increases towards $\frac{1}{2} \log 3$, the minimum viable redistribution coefficient approaches 1, and Path A becomes progressively more constrained.*

Corollary 3.3 (Double-prime window requires new methods). *For L near $\frac{1}{2} \log 3$, both the prime-2 and prime-3 terms enter the explicit formula. corollary 3.2 shows that Path A already requires θ close to 1 before prime 3 appears; including prime 3 would further tighten the constraint, making Path A unfeasible for $L > \frac{1}{2} \log 3$.*

REFERENCES

- [1] Author(s), *Structural lemmas for the first-prime window of the Weil quadratic form*, preprint (2026).
`arXiv:XXXX.XXXX`.